

Q1 → 5 $P(A|B) = \frac{5}{33}$ Conditional Probability

A, B

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

given, $P(B) \neq 0$
 $A = \{D_1 = 5\}$ $B = \{D_1 + D_2 \leq 10\}$

$$P(A \cap B) = \frac{5}{36} = \frac{5}{33}$$

D_1, D_2

$\{(1,1), (1,2) \dots (2,1), (2,2) \dots$

$(6,6)\}$ (36)

	D_1					
	1	2	3	4	5	6
1	2	3	4	5	6✓	7
2	3	4	5	6	7✓	8
3	4	5	6	7	8✓	9
4	5	6	7	8	9✓	10
5	6	7	8	9	10✓	11
6	7	8	9	10	11	12

Independent Events

A, B

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B) = P(A) \times P(B)$$

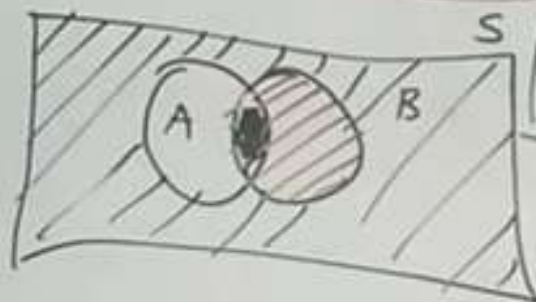
$$P(A|B) = \frac{n(A \cap B)}{n(B)}$$

(I)

$$P(A) = \frac{P(A \cap B)}{P(B)} = \frac{n(A \cap B)}{n(S) \times n(B)}$$

$$P(A) = \frac{n(A \cap B)}{n(B)}$$

(II)



$$P(A|B) = P(A)$$

Independent Events

A, B

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B) = P(A) \times P(B)$$

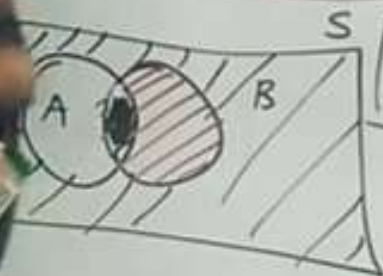
$$P(A|B) = \frac{n(A \cap B)}{n(B)}$$

①

$$P(A) = \frac{P(A \cap B)}{P(B)} = \frac{n(A \cap B)}{n(S) \times n(B)}$$

$$P(A) = \frac{n(A \cap B)}{n(B)}$$

②



$$P(A|B) = P(A)$$

Mutually Exclusive Events

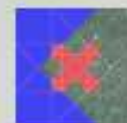
A, B

$$P(A \cap B) = 0$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A|B) = 0$$

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Bayes Theorem

↓
Naive Bayes

A, B

likelihood

→ prior

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

posterior

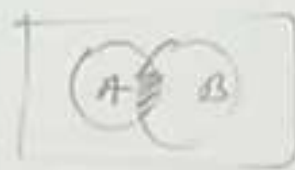
given $P(B) \neq 0$

Evidence

→ Evidence

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{--- (1)}$$

$$A \cap B = B \cap A$$



$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{P(A \cap B)}{P(A)}$$

$$P(A \cap B) = P(B|A) \times P(A) \quad \text{--- (2)}$$

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

Factory D

M_1	M_2	M_3
(20)	(30)	(50)
5%	3%	1%

Bayes Theorem

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

$$P(M_1) = \frac{1}{5} \quad P(M_2) = \frac{3}{10} \quad P(M_3) = \frac{1}{2}$$

$$P(D|M_1) = \frac{1}{20} \quad P(D|M_2) = \frac{3}{100} \quad P(D|M_3) = \frac{1}{100}$$

$$P(M_3|D) = \frac{P(D|M_3) P(M_3)}{P(D)}$$

Question

→ defective Yes

Prob M_3

$$P(D) = P(D \cap M_1) + P(D \cap M_2) + P(D \cap M_3)$$

$$P(D) = P(D|M_1)P(M_1) + P(D|M_2)P(M_2) +$$

$$P(D|M_3)P(M_3)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$\{ H=185, W=170, G=? \}$

Numerical Data In Naive Bayes

Height | Weight | Gender

172	150	M
180	170	M
165	140	M
190		M
139		F
	120	F
60	140	F
2	150	F

$$P(F | H=185, W=170) =$$

$$P(H=185 | F) P(W=170 | F) P(F)$$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$P(F)$

μ, σ

Gaussian NV

0.7

$\Rightarrow \frac{1}{2}$



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All

From the s



Naive Bayes

SENTIMENT ANALYSIS

1.5M+ VIEWS

MACHINE LEARNING

Naive Bayes Classification (Bayes Theorem)

Lec : 08

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31st MAY CODING QUESTION COMPLETE SOLUTION

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